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(d) What notes on :

(i) SONET frame format

(ii) Ring Topology 7

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# 328811(28)

## B. E. (Eighth Semester) Examination, April-May 2016

**(Old Scheme)**

**(Et & T Engg. Branch)**

## OPTICAL COMMUNICATION

***Time Allowed : Three hours***

***Maximum Marks : 80***

***Minimum Pass Marks : 28***

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*Note : Part (a) of each unit is compulsory. Attempt any two from part (b), (c) and (d) of each unit.*

### Unit-I

1. (a) Define Acceptance Angle? 2

(b) Explain step index fiber and graded index fiber with the help of suitable diagram. Also give advantages of each. 7

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(c) What is Dispersion? Discuss the different types of dispersion in optical fiber. 7

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(d) A typical relative refractive index difference for an optical fibre is 1%. Estimate numerical apertise (NA) and solid acceptance angle when core refractive index is 1·46. Also calculate critical angle at the core-cladding interface with fiber. 7

## Unit-II

2. (a) What is the spontaneous emission as applied to optical sources. 2

(b) Describe the analog modulation of LED in detail. 7

(c) Explain the principle of operation of LASER diode with the aid of suitable diagram. 7

(d) Write short notes on : 7

(i) Population Inversion

(ii) Stimulated Emission

(iii) Quantum Efficiency

## Unit-III

3. (a) Give the difference between a splice and a connector. 2

(b) What is source coupling? Describe various source coupling losses. 7

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(c) Explain the fusion splicing and mechanical splicing techniques in detail. 7

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(d) Describe the different types of star couplers with suitable diagram. 7

## Unit-IV

4. (a) What is the Responsivity? 2

(b) Explain the working principle of semiconductor junction photo diode. 7

(c) Discuss the principle of photodetection in PIN Diode. 7

(d) A PIN diode has responsivity 0·5 A/W at 0·85 $\mu$ m and 2nA of dark current. Compute the NEP (Noise Equivalent Power) as a function of the load resistance at a temperature of 300K. What is the minimum detectable power if $R_L = 100\,\Omega$ and the bandwidth is 1 MHz. 7

## Unit-V

5. (a) Define numerical aperture. 2

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(b) Explain dispersion measurement in optical fiber with the help of suitable block diagram. 7

(c) What is SONET? Discuss SONET network in detail. 7